



Society of Petroleum Engineers

**SPE-229579-MS**

## **Unleashing Cacheuta's Potential: From Mineralogy To Fractureability**

M.L. Guinazu, Universidad Nacional de Cuyo, Mendoza, Argentina / Halliburton

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229579-MS](https://doi.org/10.2118/229579-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

---

### **Abstract**

The Cuyana basin is a Triassic continental rift located in western Argentina. Formed by the deposit of thick sequences of dark shales, rich in organic matter, associated with mainly lacustrine deposits. The objective is the compositional characterization of the Cacheuta Formation, one of the three source rocks of this basin, through X-ray diffraction analysis to determine and quantify the main mineral species and evaluate the fractureability and its potential use as an unconventional shale reservoir.

Procedures, Process: To contribute to the evaluation of the Cacheuta Formation as an unconventional reservoir, fieldwork was carried out, where a geological survey and sampling of outcropping sections was carried out in classical locations of the Cuyana basin such as: Cerro Cacheuta, Cerro Cocodrilo and Divisadero Largo. The samples were described macroscopically and microscopically to understand the degree of heterogeneity (vertical and horizontal) and determine their mineralogy from direct tests such as x-ray diffraction analysis (XRD), Outcrop samples from three areas: Divisadero Largo, Cerro Cocodrilo, and Cerro Bayo, were surveyed. Seventeen samples were collected, and 12 underwent detailed XRD analysis. Based on these studies, a preliminary QFM-Clay-Calcium ternary diagram was constructed to determine the clay content of the Cacheuta Formation and hence its fractureability.

A QFM-Calcium-Clay ternary diagram indicated higher brittleness for samples from Cerro Cocodrilo and Cerro Bayo due to clay contents below 40%. Conversely, Divisadero Largo samples exhibited higher ductility due to higher clay content. The Cacheuta Formation's brittleness correlates with lower clay content (<40%), which enhances fractureability. Samples from Cerro Cocodrilo and Cerro Bayo show greater potential for hydraulic fracturing compared to Divisadero Largo. The Cacheuta Formation exhibits mineralogical characteristics suitable for hydraulic fracturing in areas with lower clay content. The identified mineralogical properties suggest brittleness and potential as an unconventional reservoir. Further geomechanical analyses are recommended to pinpoint optimal levels for stimulation.

This study provides the first mineralogical characterization of the Cacheuta Formation using XRD analysis, contributing to the understanding of its potential for hydraulic fracturing. By linking mineralogy to fractureability, this research offers a foundation for future exploration of unconventional reservoirs in the Cuyana Basin.

## Introduction

Access to energy is a key issue in any country around the world, as it promotes economic growth. For this reason, in recent times, the possibility has emerged to exploit the resources of "unconventional" reservoirs, which the country possesses in large quantities in its subsurface.

The hydrocarbons found in "unconventional" reservoirs are the same ones that have been exploited for over a century from so-called "conventional" fields. The only difference lies in the type of rock in which they are found, which implies some differences compared to traditional extraction techniques.

In the case of "shale" type reservoirs, as shown in Figure 1, the rock where the sedimentation and transformation process occurred is known as the "source rock." It is composed mostly of clay, with a small content of sand, carbonate material, and organic matter. This is the rock where hydrocarbons are hosted in "unconventional" shale oil or shale gas reservoirs.

One of the main characteristics of this source rock is its relatively low porosity and poor permeability, which means that in order to produce hydrocarbons, stimulation techniques such as hydraulic fracturing must be applied.

For these techniques to be successful, it is also necessary to determine the ductility or brittleness of the rock. Therefore, the mineralogical composition of a shale is key to achieving efficient stimulations (ductility vs. brittleness). Successful plays exhibit low clay content (<40–50%).

High quartz and/or carbonate content favors a more brittle behavior of the rock during fracturing. The lower the clay mineral content, the greater the rock's brittleness, and thus, its fracturability.

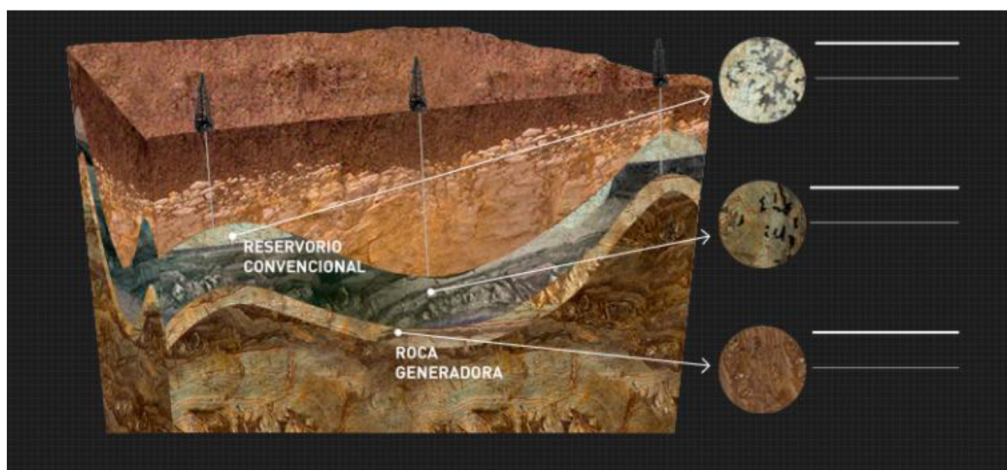


Figure 1—Reservoir Types – Source YPF

In the case of the Cuyana Basin (Figure 2), its evolution has been controlled by the structure of the basement, which influenced the development of NNW-trending normal faults that formed elongated troughs. These depocenters provided suitable conditions for the deposition of thick sequences of dark shales rich in organic matter, mainly associated with lacustrine deposits.

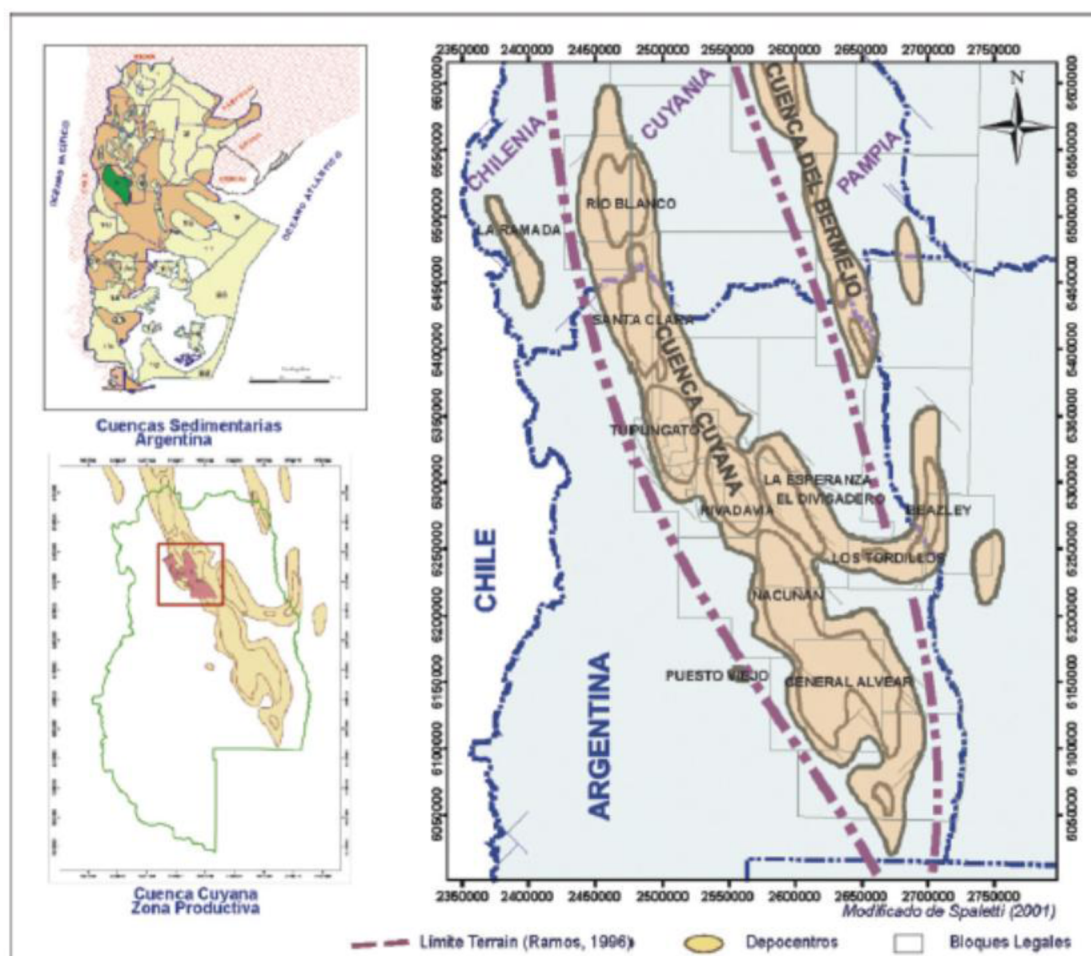


Figure 2—Location of the Cuyana Basin, indicating terrain boundaries (Ramos, 1996), depocenters

The Cuyana Basin contains three source rocks: the Cerro de las Cabras Formation, the Potrerillos Formation, and the Cacheuta Formation (Stinco and Barredo, 2014). These can be seen in the red box in Figure 3.

The Cacheuta Formation reaches variable thicknesses between 50 and 400 meters, with a Total Organic Carbon (TOC) content ranging from 3% to 10%, high thermal maturity, and Type I kerogen (Legarreta and Villar, 2011). Based on these characteristics, the Cacheuta Formation has been considered the target unit as an "unconventional" shale oil reservoir in the basin.

To contribute to the evaluation of the Cacheuta Formation as an "unconventional" reservoir, outcrop sections were surveyed and sampled in classic locations of the Cuyana Basin, such as Cerro Cocodrilo and Divisadero Largo. Their mineralogical properties were determined through X-ray diffraction (XRD) analysis, identifying the percentage of clay, quartz, and carbonate minerals, and thereby assessing the fracturability of this formation.

The main objective of this project is the compositional characterization of the sedimentary rocks that make up the Cacheuta Formation through X-ray diffraction (XRD) analysis.

Additionally, the aim is to identify and quantify the major mineral species and those included in the clay fraction.

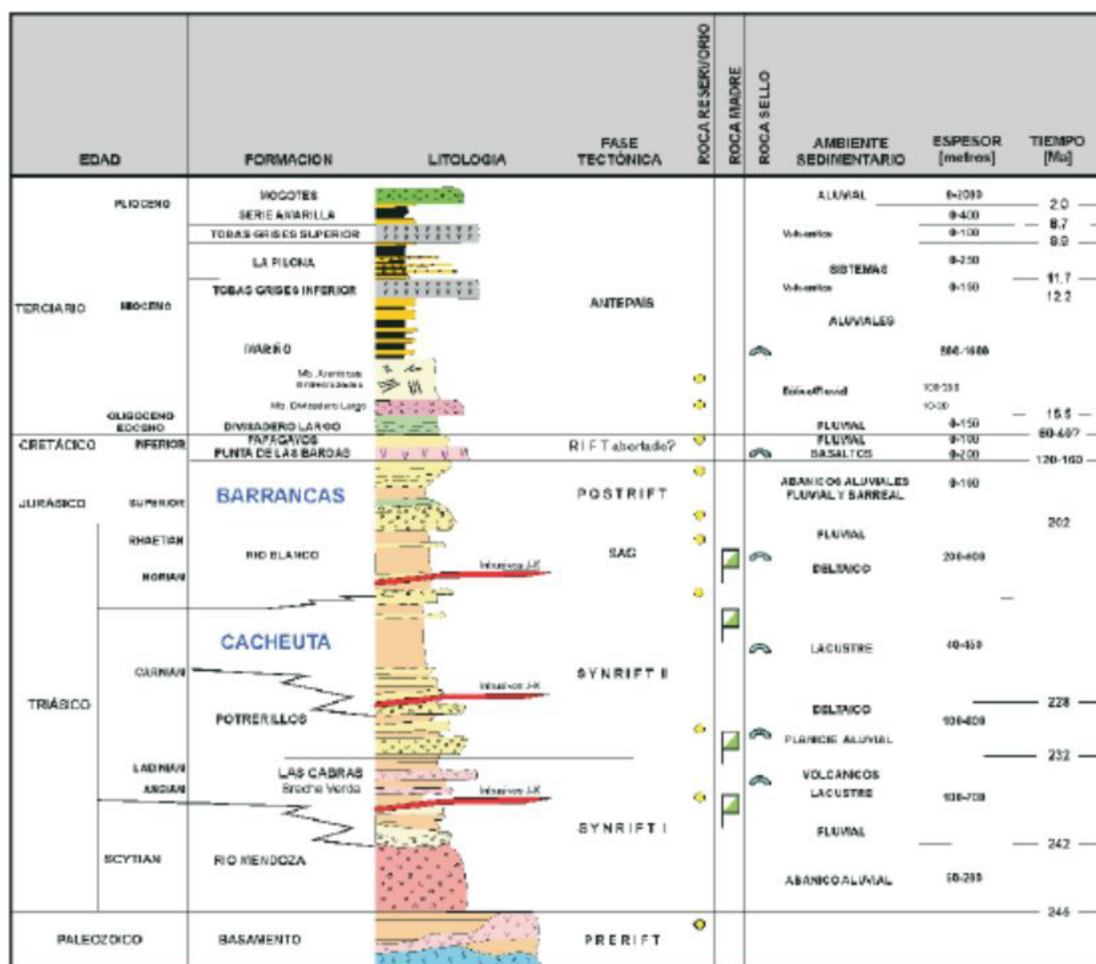


Figure 3—Chronostratigraphic column of the Cuyana Basin, modified from Kokogian et al. (1993) and Irigoyen et al. (2002). Proven and hypothetical source rocks, reservoir rocks, and seal rocks are highlighted. Cacheuta, source rock.

## Methodology

To achieve this objective, the study was conducted on samples from outcrop sections of the Cacheuta Formation in three areas: Divisadero Largo, Cerro Cocodrilo, and Cerro Bayo.

The following methodology was applied:

- Profiles were surveyed in the outcropping areas of the Cacheuta Formation in the mentioned locations.
- A total of 17 samples were collected and sent to a laboratory for diffraction studies.
- Based on these studies, a preliminary QFM–Clay–Calcium ternary diagram was created to determine the clay content of the Cacheuta Formation and, therefore, its fracturability.



Figure 4—Cacheuta in the basal third of Divisadero Largo



Figure 5—Chronostratigraphic column of the Cuyana Basin

The study of the samples detailed in Table 1, using X-ray diffraction (XRD), was carried out through total rock powder analysis (RT) and analysis of the clay fraction in oriented mounts (EO).

Table 1—Details of the samples analyzed by XRD

| SAMPLE                              | DRX |    |
|-------------------------------------|-----|----|
|                                     | RT  | EO |
| Divisadero Largo – Cacheuta Fm M-5  | X   | X  |
| Divisadero Largo – Cacheuta Fm M-6  | X   | X  |
| Divisadero Largo – Cacheuta Fm M-7  | X   | X  |
| Divisadero Largo – Cacheuta Fm M-8  | X   | X  |
| Divisadero Largo – Cacheuta Fm M-9  | X   | X  |
| Divisadero Largo – Cacheuta Fm M-10 | X   | X  |
| Cerro Cocodrilo - Cacheuta Fm M-11  | X   | X  |
| Cerro Cocodrilo - Cacheuta Fm M-12  | X   | X  |
| Cerro Cocodrilo - Cacheuta Fm M-13  | X   | X  |
| Cerro Cocodrilo - Cacheuta Fm M-14  | X   | X  |
| Cerro Cocodrilo - Cacheuta Fm M-16  | X   | X  |
| Cerro Cocodrilo - Cacheuta Fm M-17  | X   | X  |

X-ray diffraction (XRD) analyses were performed on 12 samples from the outcrop sections described above.

The primary application of X-ray Diffraction (XRD) is the qualitative identification of the mineralogical composition of a crystalline sample. It is based on optical interferences that occur when monochromatic radiation passes through a slit with a thickness comparable to the wavelength of the radiation. X-rays have wavelengths in the range of angstroms, which are on the same order as the interatomic distances within the components of crystal lattices. When irradiated onto the sample to be analyzed, the X-rays are diffracted at angles that depend on these interatomic distances.

### Sample Preparation for X-ray Diffraction

The samples were characterized by X-ray diffraction through total rock powder analysis and oriented mounts of the clay fraction.

The total rock powder analysis is performed to identify and quantify all mineral phases present in proportions greater than 5%. To carry it out, a representative sub-sample of the main sample is manually ground in an agate mortar until it reaches a particle size passing through ASTM 200 mesh, equivalent to 74  $\mu\text{m}$ . The resulting material is evenly distributed on a glass sample holder to obtain a smooth surface where most crystals are randomly oriented to be exposed to the maximum amount of X-rays. The analysis was conducted using a Rigaku SmartLab 3 X-ray diffractometer, and the data were processed with Rigaku's PDXL 2 software for both identification and quantification using the Rietveld method.

An analysis of the oriented mounts of the clay fraction is performed to identify and quantify all clay minerals present in the sample. This analysis is carried out on powder obtained from the manual disaggregation of the sample to avoid the destruction of clay minerals and the incorporation of other minerals into the fraction smaller than 10  $\mu\text{m}$ . The following steps involve the removal of calcium carbonate and calcium sulfate, centrifugation of the sample with distilled water, and pipette extraction of the suspension containing the clay fraction smaller than 2  $\mu\text{m}$ . Next, three sub-samples are prepared to differentiate between clay types. For their preparation, a one-milliliter drop of suspension is placed on a glass slide, which is then air-dried without disturbance (AD), subsequently saturated with ethylene glycol (EG), and finally calcined at 550°C (CA). The first step ensures the sedimentation of clay crystals on their basal faces once the distilled water evaporates, allowing for proper identification and quantification. The second step helps distinguish between expandable and non-expandable clays, while the third step identifies those that collapse at temperatures above 550°C.

## RESULTS OBTAINED:

As shown in [Table 1](#), 12 samples were analyzed, and for each one, a diffractogram was obtained that allowed for the quantification of minerals. Below, the results are presented, showing only the mineral quantification in summary form.

### Sample M5

Macroscopic examination with a binocular magnifying glass indicates that it is a well-consolidated shale.

The XRD analysis of the total rock indicates a composition dominated by clay minerals, followed in abundance by quartz (Qz), potassium feldspar (Fk), and plagioclase (Pg) as the main detrital minerals, accompanied by halite (Hal) and pyrite (Py), as shown in [Figure 6a](#).

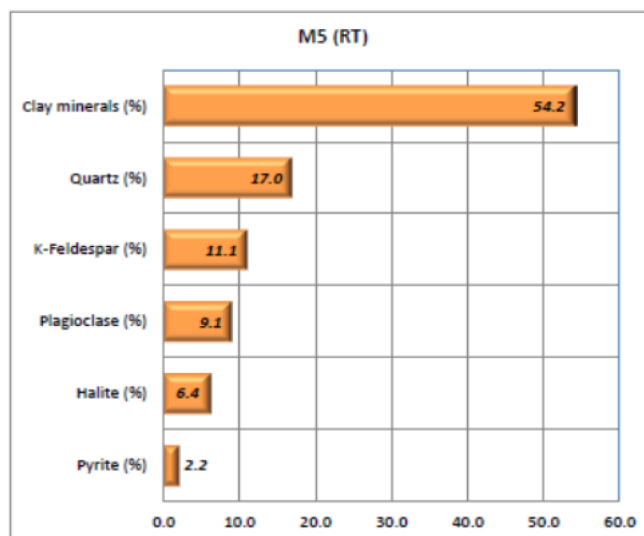


Figure 6a—Diffractogram of total rock

These detrital minerals are accompanied by a clay fraction dominated by illite (Ill), along with similar amounts of kaolinite (Kln) and smectite (Smc) (Figure 6b).

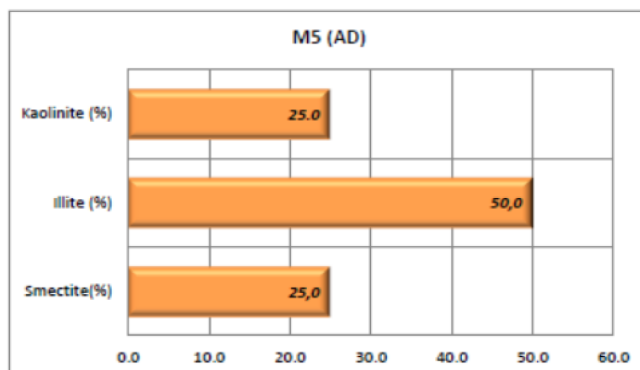


Figure 6b—Oriented mount of clay fraction

### Sample M6

Macroscopic examination with a binocular magnifying glass indicates that it is a well-consolidated shale.

XRD analysis of the total rock revealed a predominance of clay minerals (Arc), accompanied by quartz (Qz), plagioclase (Pg), and potassium feldspar (Fk) as the main detrital minerals, along with a much smaller proportion of pyrite (Py) (Figure 7a).

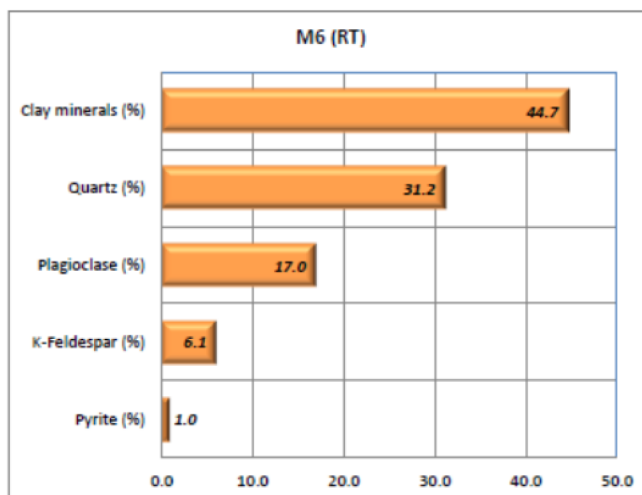


Figure 7a—Diffraction of total rock

The analysis of the clay fraction showed a marked predominance of an illite/smectite (I/S) interstratified mineral with an estimated 75% of expandable layers, followed by kaolinite (Kln) and illite (Ill) in similar proportions (Figure 7b).

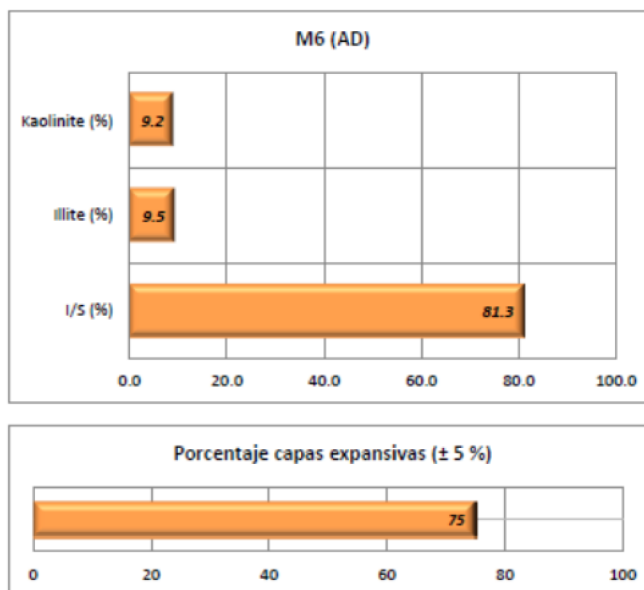


Figure 7b—Oriented amount of clay fraction

### Sample M7

Macroscopic examination with a binocular magnifying glass indicates that it is a well-consolidated shale.

XRD analysis of the total rock indicates a composition dominated by quartz (Qz) and clay minerals (Arc), accompanied by potassium feldspar (Fk) and plagioclase (Pg) (Figure 8a).

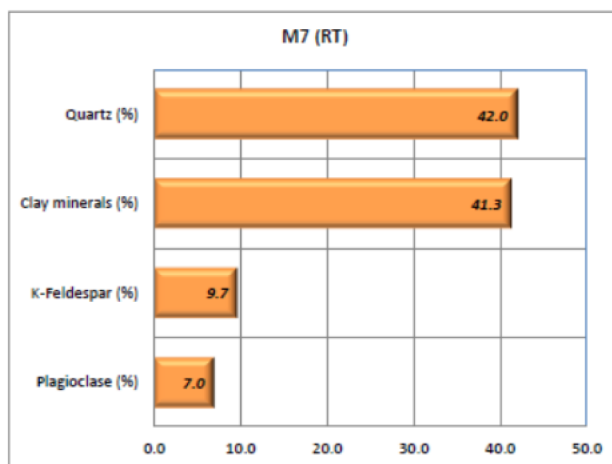


Figure 8a—Diffraction of total rock

The clay fraction shows a marked predominance of an illite/smectite (I/S) interstratified mineral with an estimated 55% of expandable layers, followed by illite (Ill) and, to a lesser extent, kaolinite (Kln) (Figure 8b).

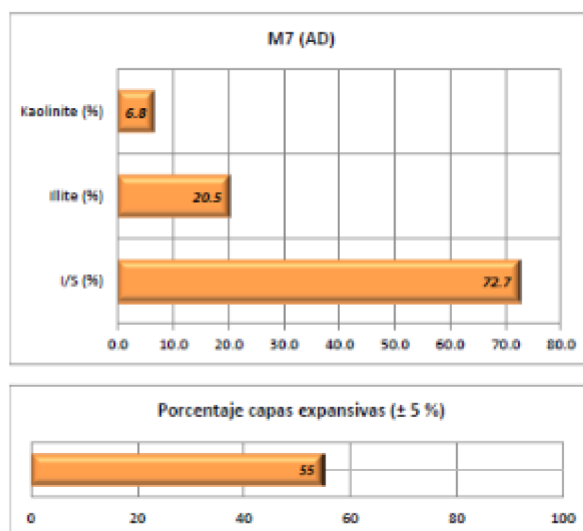


Figure 8b—Oriented mount of clay fraction

### Sample M8

Macroscopic examination with a binocular magnifying glass indicates that it is a well-consolidated shale.

XRD analysis of the total rock revealed a composition dominated by clay minerals (Arc), quartz (Qz), and potassium feldspar (Fk), accompanied by plagioclase (Pg) and a much smaller proportion of hematite (Hem) (Figure 9a).

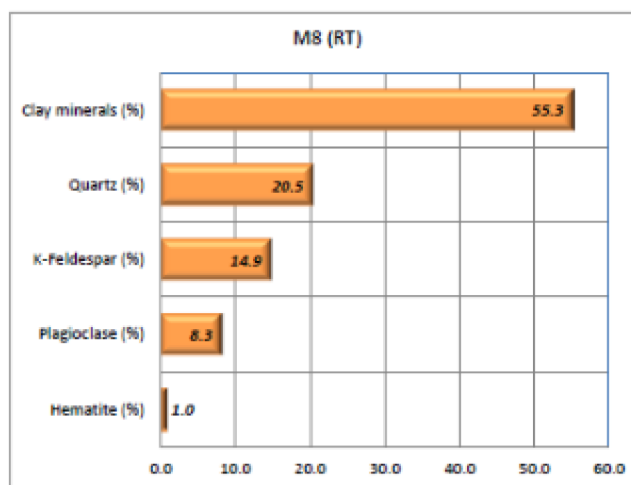


Figure 9a—Diffractogram of total rock

The clay fraction shows a predominance of kaolinite (Kln), accompanied by an illite/smectite (I/S) interstratified mineral with an estimated 35% of expandable layers, and a smaller proportion of illite (Ill) (Figure 9b).

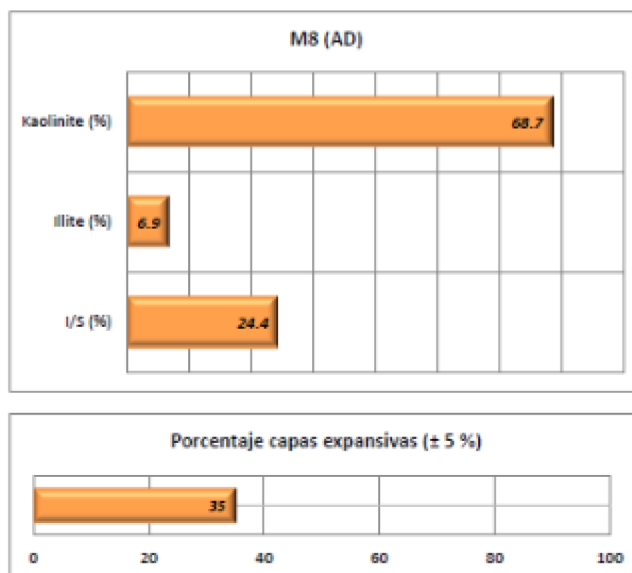


Figure 9b—Oriented mount of clay fraction

### Sample M9

Macroscopic examination with a binocular magnifying glass indicates that it is a well-consolidated shale.

XRD analysis of the total rock indicates a composition dominated by quartz (Qz) and clay minerals (Arc), followed by plagioclase (Pg) and potassium feldspar (Fk) (Figure 10a).

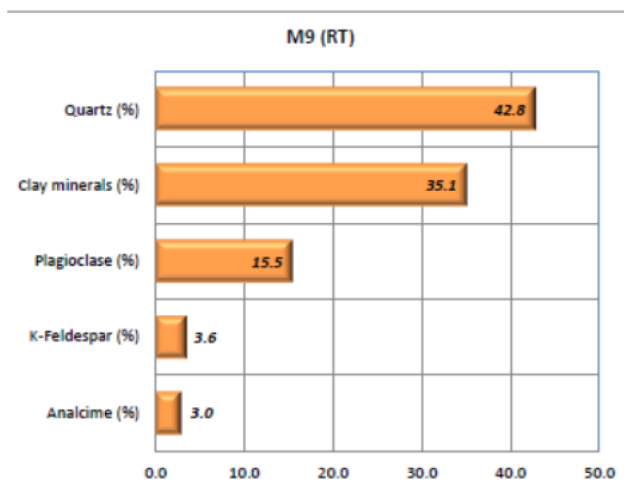


Figure 10a—Diffractogram of total rock

The analysis of the clay fraction showed a predominance of an illite/smectite (I/S) interstratified mineral with an estimated 35% of expandable layers, followed by illite (Ill) and kaolinite (Kln) (Figure 10b).

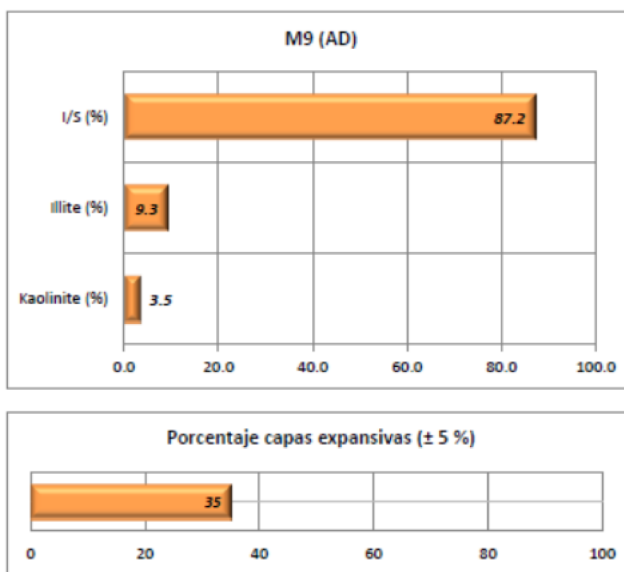


Figure 10b—Oriented mount of clay fraction

### Sample M10

Macroscopic examination with a binocular magnifying glass indicates that it is a well-consolidated shale.

XRD analysis of the total rock revealed a predominance of quartz (Qz), potassium feldspar (Fk), and muscovite (Ms), accompanied to a lesser extent by plagioclase (Pg) as the main detrital minerals, along with a smaller proportion of clay minerals (Arc) (Figure 11a).

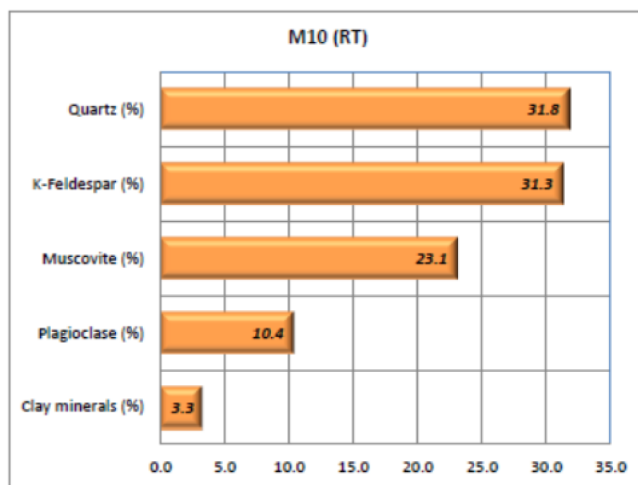


Figure 11a—Diffractogram of total rock

The analysis of the clay fraction shows a predominant presence of an illite/smectite (I/S) interstratified mineral with an estimated 45% of expandable layers, followed by illite (Ill) and kaolinite (Kln) (Figure 11b).

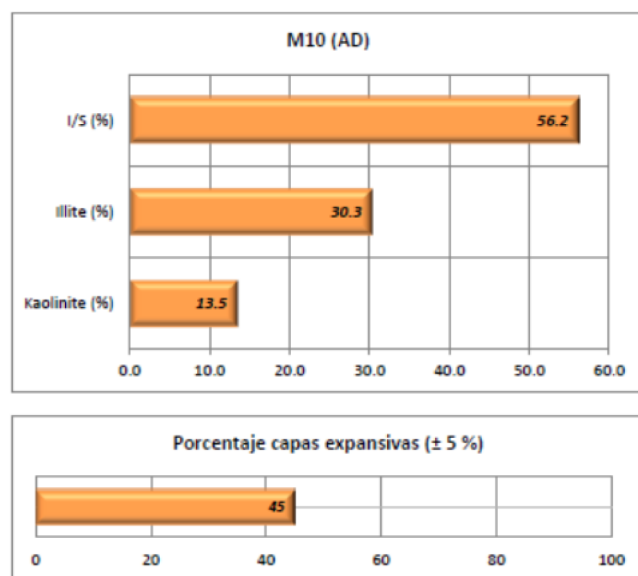


Figure 11b—Oriented mount of clay fraction

### Sample M11

Macroscopic examination with a binocular magnifying glass indicates that it is a medium to coarse sandstone, with a quartz-rich composition and a clay matrix, showing poor sorting.

XRD analysis of the total rock detected a predominance of quartz (Qz) and clay minerals (Arc) as the most abundant mineral phases, followed by potassium feldspar (Fk) and, to a lesser extent, plagioclase (Pg) (Figure 12a).

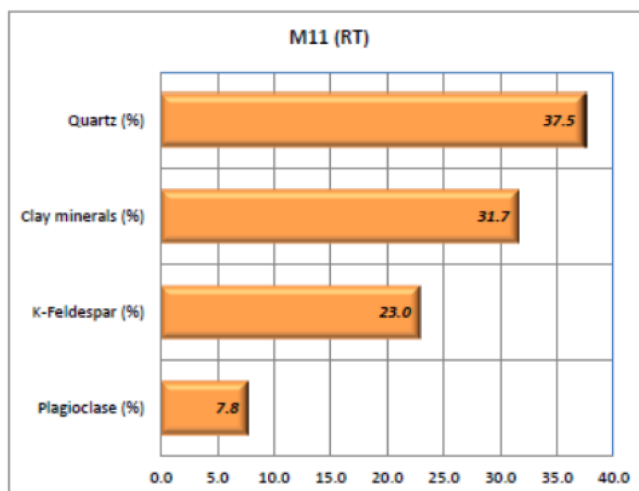


Figure 12a—Diffractogram of total rock

The clay fraction determined by XRD indicates a marked dominance of kaolinite (Kln), followed by an illite/smectite (I/S) interstratified mineral with an estimated 35% of expandable layers, and a smaller amount of illite (Ill) (Figure 12b)

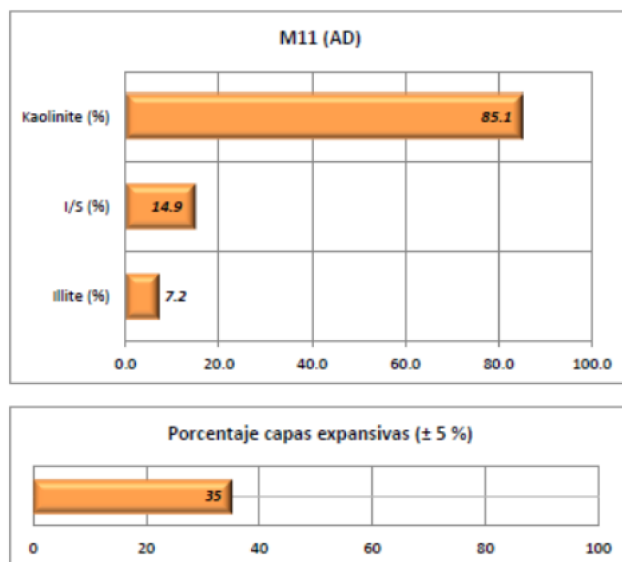


Figure 12b—Oriented mount of clay fraction

### Sample M12

Macroscopic examination with a binocular magnifying glass indicates that it is a well-consolidated shale.

XRD analysis of the total rock revealed a predominance of quartz (Qz) and clay minerals (Arc), followed by potassium feldspar (Fk) and plagioclase (Pg) (Figure 13a).

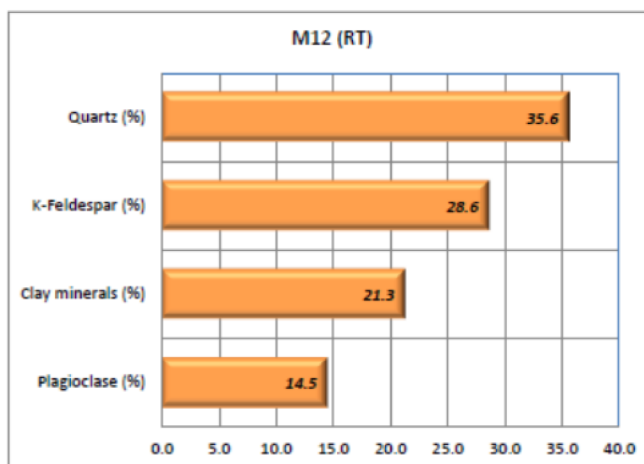


Figure 13a—Diffractogram of total rock

The analysis of the clay fraction shows a predominant presence of illite (Ill), followed by an illite/smectite (I/S) interstratified mineral with an estimated 80% of expandable layers, and a smaller proportion of kaolinite (Kln) (Figure 13b)

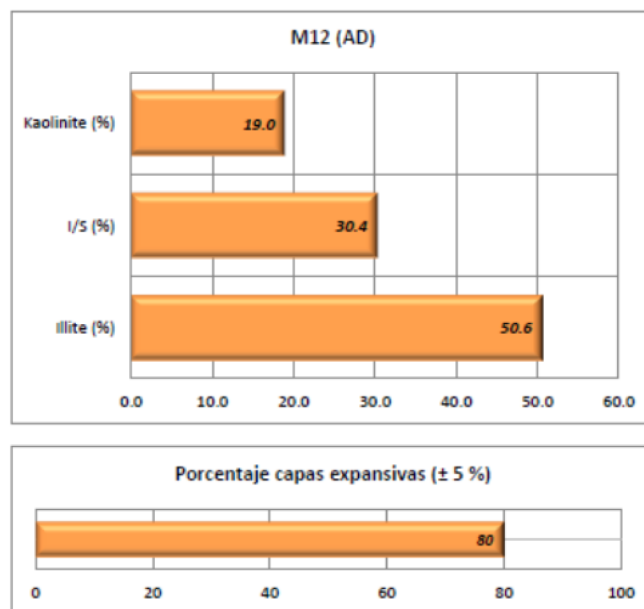


Figure 13b—Oriented mount of clay fraction

### Sample M13

Macroscopic examination with a binocular magnifying glass indicates that it is a well-consolidated shale.

XRD analysis of the total rock indicates a predominance of quartz (Qz), along with potassium feldspar (Fk) as the main detrital minerals, accompanied by clay minerals (Arc) and a smaller proportion of plagioclase (Pg) (Figure 14a).

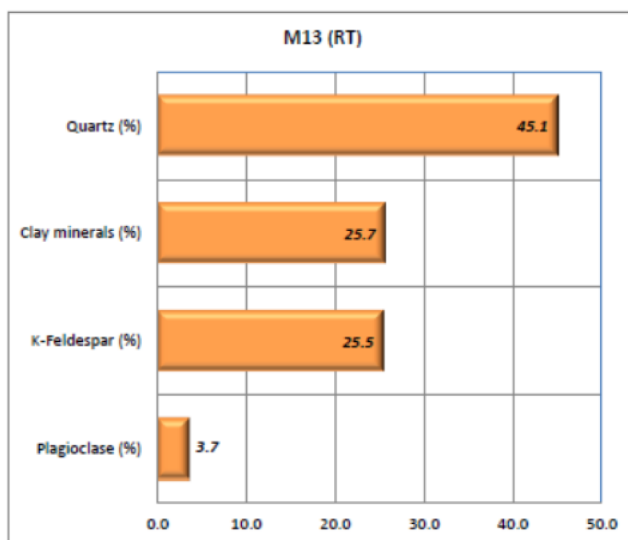


Figure 14a—Diffractogram of total rock

The clay fraction shows a composition dominated by smectite (Smc), followed by kaolinite (Kln) and illite (Ill) in similar proportions (Figure 14b).

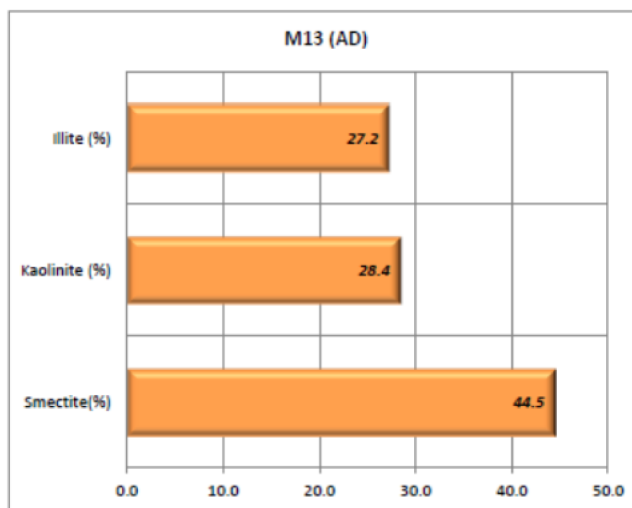


Figure 14b—Oriented mount of clay fraction

### Sample M14

Macroscopic examination with a binocular magnifying glass indicates that it is a well-consolidated pelite.

X-ray diffraction (XRD) of the whole rock reveals a predominance of quartz (Qz) and clay minerals (Clays), accompanied by potassium feldspar (Kfs) and plagioclase (Plg) in lesser amounts. (Figure 15a)

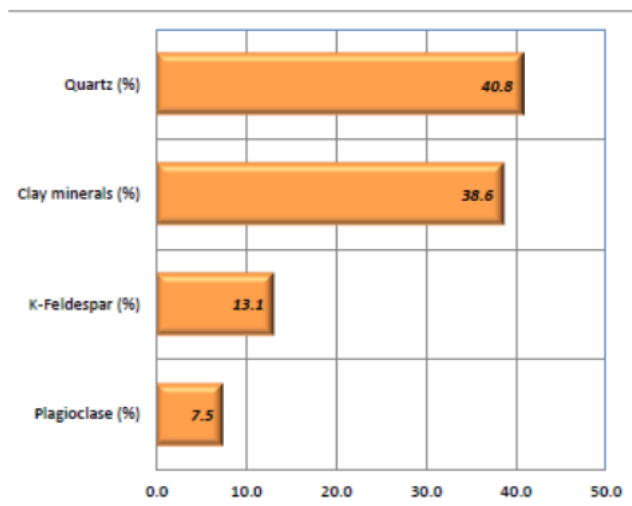


Figure 15a—Diffractogram of total rock

The clay fraction shows a marked dominance of smectite (Smc), followed by illite (Ill) and a minor proportion of kaolinite (Kln). (Figure 15b)

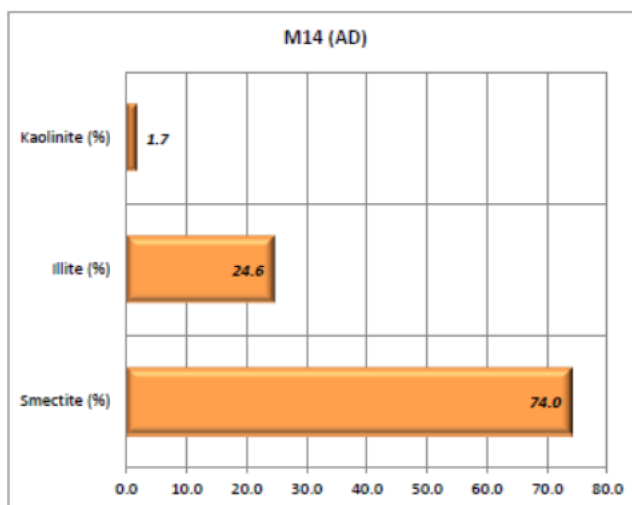


Figure 15b—Oriented mount of clay fraction

### Sample M16

Macroscopic examination with a binocular magnifying glass indicates that it is a well-consolidated pelite.

X-ray diffraction (XRD) of the whole rock reveals a predominance of quartz (Qz), followed by clay minerals (Clays), and potassium feldspar (Kfs) and plagioclase (Plg) in similar proportions. (Figure 16a)

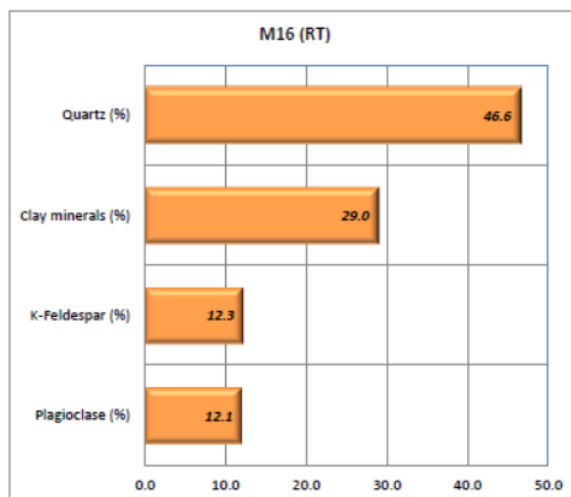


Figure 16a—Diffraction of total rock

The clay fraction of this sample is completely dominated by kaolinite (Kln). (Figure 16b)

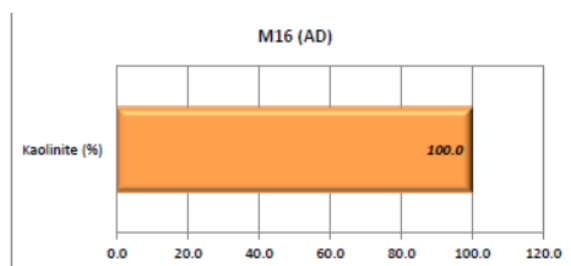


Figure 16b—Oriented mount of clay fraction

### Sample M17

Macroscopic examination with a binocular magnifying glass indicates that it is a well-consolidated pelite.

X-ray diffraction (XRD) of the whole rock revealed a predominance of clay minerals (Clays), accompanied by quartz (Qz), potassium feldspar (Kfs), muscovite (Ms), and plagioclase (Plg). (Figure 17a)

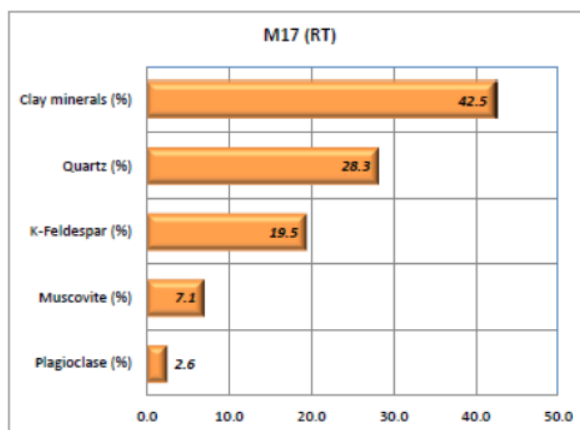


Figure 17a—Diffraction of total rock

The clay fraction is dominated by an illite/smectite (I/S) interstratified mineral with an estimated percentage of expandable layers of 35%, followed by kaolinite (Kln) and illite (Ill). (Figure 17b)

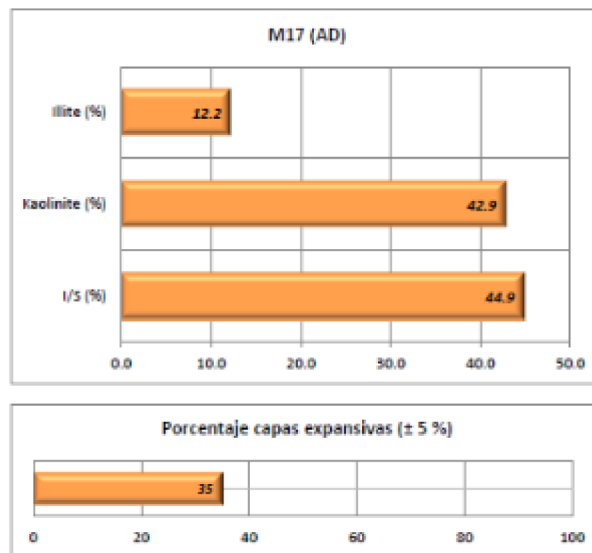


Figure 17b—Oriented mount of clay fraction

Based on the results, a ternary diagram QFM–Calcium–Clay was constructed. (Figure 18) Knowing that successful plays are those with low clay content (below 40%), as they exhibit greater brittleness, the diagram highlights the samples that meet this criterion. These correspond to those collected from Cerro Cocodrilo and Cerro Bayo. In contrast, the samples from Divisadero Largo show higher clay content, which makes the formation more ductile.

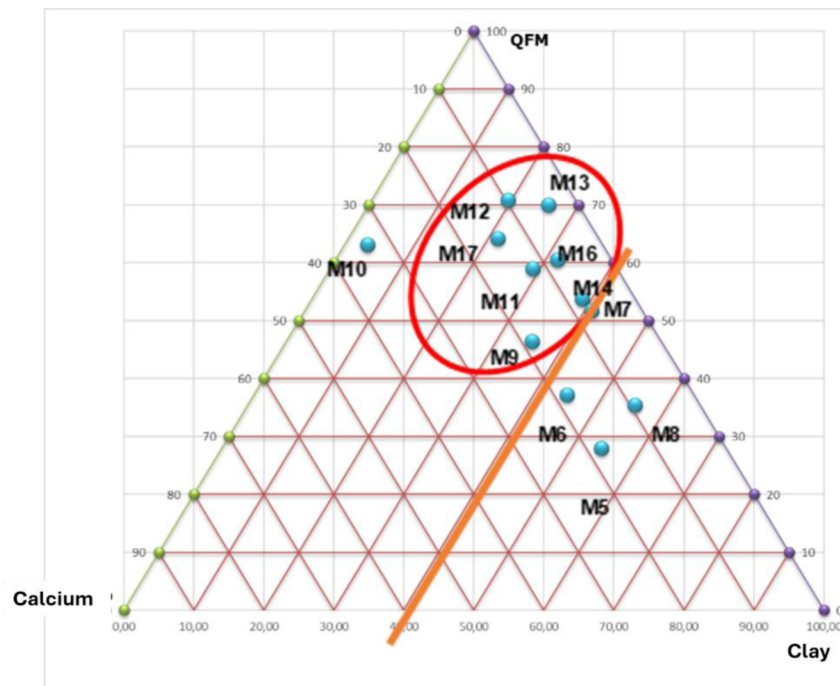


Figure 18—QFM–Calcium–Clay diagram

## Conclusions

The analysis of the samples through X-ray diffraction (XRD) showed a well-consolidated pelite, where the clay phase is composed of illite, kaolinite, and smectite.

The results have allowed for the characterization of the mineralogical properties of the Cacheuta Formation. The clay percentages in the samples from Cerro Bayo and Cerro Cocodrilo indicate that the Cacheuta Formation exhibits brittle behavior, as they contain less than 40% clay.

Since the efficiency of hydraulic stimulation is key to shale productivity and is proportional to the rock's fracturability, we can infer that the Cacheuta Formation will be a fracturable formation.

It is recommended to complement this study with geomechanical analyses. This will allow for the identification of promising intervals for hydraulic fracturing and thus highlight the potential of the Cacheuta Formation as an unconventional reservoir.

## Acknowledgments

The author would like to express sincere gratitude to the Faculty of Engineering at the National University of Cuyo for the opportunity to carry out this research project, and to the Geology Laboratory at the National University of Río Negro for their invaluable support and collaboration throughout the analytical work. Their contributions were essential to the development of this study and are deeply appreciated.

## References

- Stinco, L., & Barredo, S. (2014). Geological characteristics and resources associated with unconventional shale-type reservoirs in Argentina's productive basins.
- Stinco, L., Sosa Massaro, A., Abarzúa, F., Fuenmayor, E., & Abalos. (2016). Depositional controls over the lacustrine source rocks of the Cuyana Basin.
- Zencich, Villar, & Boggetti, D. (2008). Cacheuta–Barrancas Petroleum System of the Cuyana Basin.
- Boggetti, D., Scolari, J. C., & Regazzoni, C. (2002). Cuyana Basin: Geological framework and historical overview of petroleum activity.
- Barredo, S., & Stinco, L. (2010). Geodynamics of sedimentary basins: Their importance in locating petroleum systems in Argentina.
- Barredo, S., & Stinco, L. (2013). A Geodynamic View of Oil and Gas Resources Associated with Unconventional Shale Reservoirs in Argentina.
- Legarreta, L., & Villar, H. (2011). Geological and Geochemical Keys of the Potential Shale Resources.